

Practical No. 1

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#include <iostream>
#include <algorithm>
using namespace std;
struct Student
{
    int
    id;
    string name;
    float cgpa;
};
int linearSearch (Student s[], int n, int key)
{
    for(int i=0; i<n; i++)
    {
        if(s[i] . id==key)
        return i;
    }
    return -1;
}
int binarySearch(Student s[], int n, int key)
{
    int low=0;    int
    high=n-1;
    while(low<=high)
    {
        int mid=(low+high)/2;
        if(s[mid]
        .id==key)
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return mid;    if
(key<s[mid] .id)
high=mid-1;    else
low=mid+1;
    }
return -1;
}

```

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void bubbleSortName(Student s[] , int n)
{
    for(int i=0;i<n-1;i++)
    {
        for(int j=0;j<n-i-1;j++)
        {
            if(s[j] .name>s[j+1] .name)
swap(s[j],s[j+1]);
        }
    }
}

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void selectionSortCGPA(Student s[], int n)
{
    for(int i=0;i<n-1;i++)
    {
        int min=i;
        for(int j=i+1;j<n;j++)
        {
            if(s[j].cgpa<s[min].cgpa)
                min=j;
        }
        swap(s[i],s[min]);
    }
}

```

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    }
}

int main()
{
    int
n;

    cout<<"Enter Number of Students :";
cin>>n;

    Student *s=new Student [n];
for(int i=0;i<n;i++)
{
    cout<<"\nStudent "<< i+1 << endl ;
cout<<"ID : ";    cin>>s[i] .id;
cout<<"Name :";    cin>>s[i] . name;
cout<<"CGPA :";    cin>>s[i] .cgpa;

}
cout<<"\nStudent Records\n"; for(int
i=0;i<n;i++)
{
    cout << s[i] .id << " "
    << s[i] .name << " "
    << s[i] .cgpa << endl;
}

int key; cout<<"\nEnter ID to Search :";
cin>>key; int pos=linearSearch(s,n,key);
if(pos!=-1)    cout<<"Found using Linear
search" << endl; else    cout<<"Not Found" <<
endl; bubbleSortName(s,n); cout<<"\nSorted by
Name\n"; for(int i=0;i<n;i++)

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{
    cout << s[i]. id << " "
        << s[i]. name << " "
        << s[i]. cgpa << endl;
}
selectionSortCGPA(s,n);
cout<<"\nSorted by CGPA\n"; for(int
i=0;i<n;i++)
{
    cout << s[i]. id << " "
        << s[i]. name << " "
        << s[i]. cgpa << endl;
}
sort(s, s + n, [] (Student a,Student b)
{
    return a.id < b. id;
});
pos=binarySearch(s,n,key); if(pos!=-1)    cout <<
"\nFound using Binary Search " << endl; else
cout << "\nNot Found" << endl; delete[] s; return 0;
}

```

Output:

Enter Number of Students : 2

Student 1

ID : 115

Name :Pranjali

CGPA :8.89

Student 2

ID : 120

Name :Riddhi

CGPA :8.73

Student Records

115 Pranjali 8.89

120 Riddhi 8.73

Enter ID to Search :115

Found using Linear search

Sorted by Name

115 Pranjali 8.89

120 Riddhi 8.73

Sorted by CGPA

120 Riddhi 8.73

115 Pranjali 8.89

Found using Binary Search

=== Code Execution Successful ===